

The evolution of the supercontinent Gondwana represents a clear path in Earth's cycle. It can serve as a perfect example of how the planet shifted from the unstable "proto-continent era" of the Archean to the rigid plate tectonics in the Phanerozoic. The evolution of Gondwana wasn't a single event but a long series of orogenic sutures that first started in the Late Neoproterozoic. 2.5 billion years ago, before Gondwana could form, the Earth's crust went through a rheological hardening at the Archean-Proterozoic boundary. Later, as the mantle cooled, the soft vertical tectonics and diapirism (as seen in Archean greenstone-granite belts (Windley, 1995, Chap. 19)) were replaced by the linear, rigid plate movements (as seen in Early Proterozoic orogens (Windley, 1995, Chap. 16)). This stabilization means the crust could hold onto massive passive margins and transmit horizontal stress across thousands of kilometers (Hoffman, 2003). Gondwana was built by closing several huge ocean basins, such as the Brasiliano and Pan-African oceans. This led to the Pan-African and Brasiliano orogenies, which were around 600 - 500 Ma (Unrug, 1996). The sutures in the rock can still be seen today through ophiolitic sequences, blueschist-facies metamorphism, and the change from deep-water flysch to terrestrial molasse sediments. This progress was recorded by paleomagnetic data, by looking at Apparent Polar Wander (APW) paths, scientists have found that the southern continents (Africa, South America, Antarctica, Australia, and India) moved together as one block for most of the Paleozoic (Ramos, 2008). The internal structure of the supercontinent was held together by high-grade granulite-gneiss belts. 180 million years ago, in the Mid-Jurassic, Gondwana began to break apart. This can be tracked through oceanic magnetic anomaly patterns and the way APW paths begin to move away from each other (Lawver & Scotese, 1987). This breakup started with massive mantle-plume activity and mafic dyke swarms, as similar to the patterns seen earlier in the Proterozoic (Windley, 1995, Chap. 16). The final split of Antarctica from South America and Australia (45 - 30 Ma) changed the global climate forever, transformed it from a warm "greenhouse" to a cold "icehouse" climate. In conclusion, Gondwana's history shows how the Earth's cooling controls the mechanical life cycle of the continental crust.

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